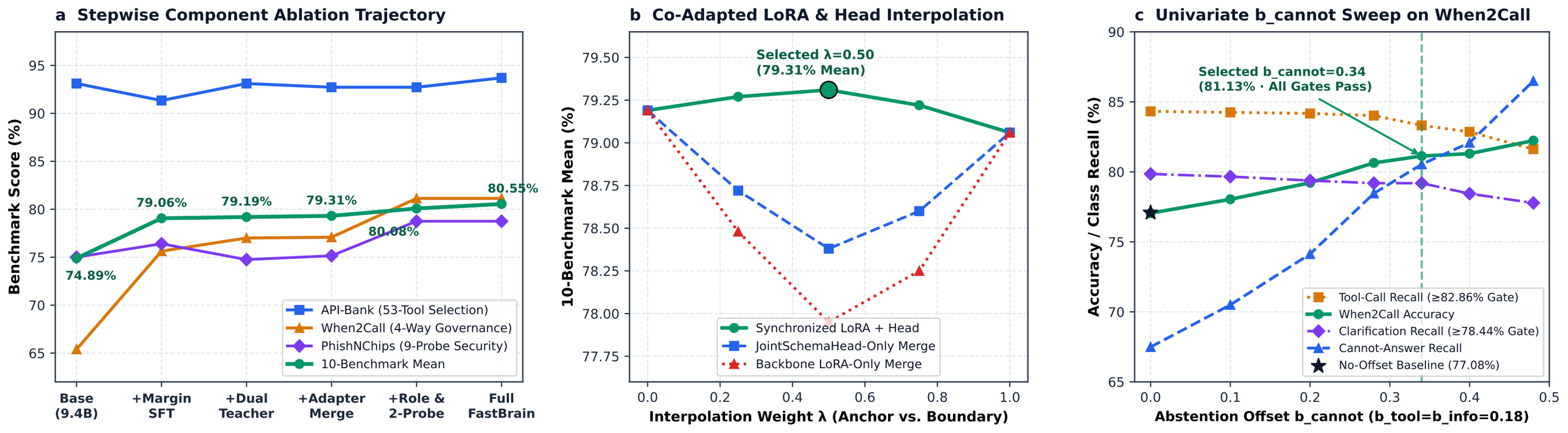
a Stepwise Component Ablation Trajectory
Benchmark Score (%)
95
90
85
80
75
70
65
74.89%
79.06%
79.19%
79.31%
80.08%
80.55%
API-Bank (53-Tool Selection)
When2Call (4-Way Governance)
PhishNChips (9-Probe Security)
10-Benchmark Mean
Base (9.4B)
+Margin SFT
+Dual Teacher
+Adapter Merge
+Role & 2-Probe
Full FastBrain
b Co-Adapted LoRA & Head Interpolation
10-Benchmark Mean (%)
79.50
79.25
79.00
78.75
78.50
78.25
78.00
77.75
Selected λ=0.50
(79.31% Mean)
Synchronized LoRA + Head
JointSchemaHead-Only Merge
Backbone LoRA-Only Merge
0.0
0.2
0.4
0.6
0.8
1.0
Interpolation Weight λ (Anchor vs. Boundary)
c Univariate b_cannot Sweep on When2Call
Accuracy / Class Recall (%)
90
85
80
75
70
65
Selected b_cannot=0.34
(81.13% · All Gates Pass)
Tool-Call Recall (≥82.86% Gate)
When2Call Accuracy
Clarification Recall (≥78.44% Gate)
Cannot-Answer Recall
No-Offset Baseline (77.08%)
0.0
0.1
0.2
0.3
0.4
0.5
Abstention Offset b_cannot (b_tool=b_info=0.18)